Turcsán István – HC7Q7Q

Felhasználói dokumentáció

A Snake Game játékhoz

Az alkalmazást futtatva a főmenübe érkezünk meg, ahol 3 lehetséges opció közül választhatunk:

- PLAY, avagy egy játék elindítása
- SCORES, avagy a ranglista megtekintése
- EXIT, avagy kilépés az alkalmazásból

Ha a felhasználó az első opciót választja, rögtön elindul egy játék.

Játék menete

A játék egy 11x11-es rácson zajlik. A játékos egy kígyót irányít, amely folyamatosan halad előre.

Cél: megenni a piros almákat. Minden alma elfogyasztása növeli a kígyó hosszát és a pontszámot. Megenni egy almát úgy lehet, hogy a játékos átmegy rajta.

Játék vége: ha a kígyó falnak ütközik, vagy belemegy a saját testébe

Győzelem: ha a kígyó betölti a teljes pályát (121 hosszú lesz)

Irányítás

A játék közben a W A S D billentyűzetekkel lehet irányítani a kígyót, mely rendre fel, balra, le és jobbra mozgás. A SPACE gombbal lehet a felvett Power Up-ot aktiválni

Power Upok

A játékban minden 10. alma megevése után egy különleges alma jelenik meg, mely lehet:

- Lila alma: Speed Power Up, aktiváláskor a kígyó 8 mp-ig kétszer olyan gyorsan megy, és az ezidő alatt megevett almák 2x annyi pontot érnek
- Fehér alma: Inverted Controls Power Up, aktiváláskor megfordulnak az irányítások 8 mp-ig, ezalatt minden megevett alma 10 pontot ér
- Sötét alma: Mega Apple Power Up, aktiváláskor a következő 8 másodpercben megjelenő almák extra nagyok lesznek, és 4 pontot érnek

Játék vége

Miután megnyertük a játékot, vagy vesztettünk, választhatunk:

- új játékot indítunk: ekkor egy teljesen új játék kezdődik azonnal
- visszalépünk a főmenübe

- elmentjük az eredményünket, melyet nevünk megadásával tehetünk meg
- kilépünk a játékból

Snake Game

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Chapter 2

Namespace Index

2.1 Package List

Here are the packages with brief descriptions (if available):

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nhf.core 13
nhf.gui 13

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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  nhf.gui.GamePanel.InGamePanel 20
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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nhf.gui.GamePanel 19
nhf.gui.GamePanel.InGamePanel 20
nhf.Main 22
nhf.core.Map 22
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Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

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Chapter 6

Directory Documentation

6.1 snakegame/src/main/java/nhf/core Directory Reference

Files

- file Engine.java
- file Map.java
- file ObjectType.java
- file Scores.java
- file Snake.java

6.2 snakegame/src/main/java/nhf/gui Directory Reference

Files

- file GamePanel.java
- file ScorePanel.java

6.3 snakegame/src/main/java Directory Reference

Directories

- directory nhf

6.4 snakegame/src/main Directory Reference

Directories

- directory java

6.5 snakegame/src/main/java/nhf Directory Reference

Directories

- directory core
- directory gui

Files

- file Main.java

6.6 snakegame Directory Reference

Directories

- directory src

6.7 snakegame/src Directory Reference

Directories

- directory main

Chapter 7

Namespace Documentation

7.1 Package nhf

Packages

- package core
- package gui

Classes

- class Main

7.1.1 Detailed Description

Here is a snapshot of my new application: uml.png "UML Diagram"

7.2 Package nhf.core

Classes

- class Engine
- class Map
- enum ObjectType
- class Scores
- class Snake

7.3 Package nhf.gui

Classes

- class GamePanel
- class ScorePanel

Chapter 8

Class Documentation

8.1 nhf.core.Engine Class Reference

Public Member Functions

- Engine (InGamePanel igp)
- void gameEnded (String message)
- void resetGame ()
- boolean headInBody (Point head, Snake snake)
- void loop ()
- String getFormattedTime ()
- void deleteObjectFromMap (ObjectType type)
- ObjectType getEatenObjectType (Point nextHead)
- void generatePowerUp ()
- void generateApple ()
- void start ()
- void changeDirection (int keyCode)
- boolean isOccupied (int x, int y)
- void activatePowerUp ()
- ObjectType getStoredPowerUp ()
- int getTimeLeft ()
- boolean isMegaAppleModeOn ()
- boolean currentAppleMega ()

8.1.1 Detailed Description

Implements the main game logic

8.1.2 Constructor & Destructor Documentation

8.1.2.1 Engine()

```
nhf.core.Engine.Engine (
            InGamePanel igp)
```

Constructor, sets up the engine: new map, new snake, initial direction etc.

Parameters

igp	

8.1.3 Member Function Documentation

8.1.3.1 activatePowerUp()

```
void nhf.core.Engine.activatePowerUp ()
```

Activates the stored power-up and applies its effect

8.1.3.2 changeDirection()

```
void nhf.core.Engine.changeDirection (
            int keyCode)
```

Changes the snake's direction based on user input, considers inverted controls if the power-up is active

Parameters

keyCode	

8.1.3.3 currentAppleMega()

```
boolean nhf.core.Engine.currentAppleMega ()
```

Checks if the current apple is mega

Returns

wheter the current apple is mega

8.1.3.4 deleteObjectFromMap()

```
void nhf.core.Engine.deleteObjectFromMap (
            ObjectType type)
```

Removes the power-up from the map after being picked up

Parameters

type	

8.1.3.5 gameEnded()

```
void nhf.core.Engine.gameEnded (
            String message)
```

Ends the game with either a win or a loss message. Gives the user options to start a new game, return to menu, save their score or exit from the game.

Parameters

message	

8.1.3.6 generateApple()

```
void nhf.core.Engine.generateApple ()
```

Generates a new apple at a random position on the map that is free (there is no snake body part or power-up there)

8.1.3.7 generatePowerUp()

```
void nhf.core.Engine.generatePowerUp ()
```

Generates a power-up at a random position on the map that is free (there is no snake body part or apple there)

8.1.3.8 getEatenObjectType()

```
ObjectType nhf.core.Engine.getEatenObjectType (
            Point nextHead)
```

Determines what type of object the snake has eaten based on its next head position

Parameters

nextHead	

Returns

8.1.3.9 getFormattedTime()

```
String nhf.core.Engine.getFormattedTime ()
```

Gets the formatted time since the start of the game in mm:ss format

8.1.3.10 getStoredPowerUp()

```
ObjectType nhf.core.Engine.getStoredPowerUp ()
```

Gets the currently stored power-up

Returns

the stored power-up

8.1.3.11 getTimeLeft()

```
int nhf.core.Engine.getTimeLeft ()
```

Gets the time left for the currently stored power-up

Returns

the time left in seconds

8.1.3.12 headInBody()

```
boolean nhf.core.Engine.headInBody (
            Point head,
            Snake snake)
```

Checks if the snake's head collides with its body. Skips the first point as it's the head itself

Parameters

head	
snake	

Returns

wheter head collides with body

8.1.3.13 isMegaAppleModeOn()

```
boolean nhf.core.Engine.isMegaAppleModeOn ()
```

Checks if mega apple mode is currently active

Returns

wheter mega apple mode is on

8.1.3.14 isOccupied()

```
boolean nhf.core.Engine.isOccupied (
            int x,
            int y)
```

Checks if a given position on the map is occupied by the snake, apple or power-up

Parameters

x	
y	

Returns

wheter the position is occupied

8.1.3.15 loop()

```
void nhf.core.Engine.loop ()
```

Game loop, that is called on every tick

8.1.3.16 resetGame()

```
void nhf.core.Engine.resetGame ()
```

Resets the game to its initial state

8.1.3.17 start()

```
void nhf.core.Engine.start ()
```

Starts the game loop timer

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/core/Engine.java

8.2 nhf.gui.GamePanel Class Reference

Classes

- class MenuPanel
- class InGamePanel

Public Member Functions

- GamePanel ()
- void menu ()

8.2.1 Detailed Description

The main game panel that handles the menu and in-game display

8.2.2 Constructor & Destructor Documentation

8.2.2.1 GamePanel()

```
nhf.gui.GamePanel.GamePanel ()
```

Constructs the GamePanel and initializes the menu

8.2.3 Member Function Documentation

8.2.3.1 menu()

```
void nhf.gui.GamePanel.menu ()
```

Sets up and displays the main menu

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/gui/GamePanel.java

8.3 nhf.gui.GamePanel.InGamePanel Class Reference

Public Member Functions

- void setSnake (Snake snake)
- void setMap (Map map)
- void setEngine (Engine engine)
- void updateGameStatus (ObjectType item, int time, int score, String elapsedTime, int dx, int dy)
- void paintComponent (Graphics g)
- void paintObjects (Point object, int offsetX, int offsetY, int cellSize, Graphics g, Color color)
- void returnToMenu ()

8.3.1 Detailed Description

The panel displayed during the game

8.3.2 Member Function Documentation

8.3.2.1 paintComponent()

```
void nhf.gui.GamePanel.InGamePanel.paintComponent (
            Graphics g)
```

8.3.2.2 paintObjects()

```
void nhf.gui.GamePanel.InGamePanel.paintObjects (
            Point object,
            int offsetX,
            int offsetY,
            int cellSize,
            Graphics g,
            Color color)
```

Paints game objects like power-ups

Parameters

object	
offsetX	
offsetY	
cellSize	
g	
color	

8.3.2.3 returnToMenu()

```
void nhf.gui.GamePanel.InGamePanel.returnToMenu ()
```

8.3.2.4 setEngine()

```
void nhf.gui.GamePanel.InGamePanel.setEngine (
            Engine engine)
```

8.3.2.5 setMap()

```
void nhf.gui.GamePanel.InGamePanel.setMap (
            Map map)
```

8.3.2.6 setSnake()

```
void nhf.gui.GamePanel.InGamePanel.setSnake (
            Snake snake)
```

8.3.2.7 updateGameStatus()

```
void nhf.gui.GamePanel.InGamePanel.updateGameStatus (
            ObjectType item,
            int time,
            int score,
            String elapsedTime,
            int dx,
            int dy)
```

Updates the game status display

Parameters

item	
time	
score	

elapsedTime	

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/gui/GamePanel.java

8.4 nhf.Main Class Reference

Static Public Member Functions

- static void main (String[ ] args)

8.4.1 Member Function Documentation

8.4.1.1 main()

```
void nhf.Main.main (
             String[ ] args)  [static]
```

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/Main.java

8.5 nhf.core.Map Class Reference

Public Member Functions

- Map ()
- Point getApple ()
- Point getPowerUpLocation ()
- ObjectType getPowerUpType ()
- void setApple (Point p)
- void setPowerUp (Point p, ObjectType ot)
- int[ ][ ] getGrid ()

Static Public Attributes

- static final int SIZE = 11

8.5.1 Detailed Description

The game map represented as a 2 dimensional grid.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 Map()

```
nhf.core.Map.Map ()
```

8.5.3 Member Function Documentation

8.5.3.1 getApple()

```
Point nhf.core.Map.getApple ()
```

8.5.3.2 getGrid()

```
int[][] nhf.core.Map.getGrid ()
```

8.5.3.3 getPowerUpLocation()

```
Point nhf.core.Map.getPowerUpLocation ()
```

8.5.3.4 getPowerUpType()

```
ObjectType nhf.core.Map.getPowerUpType ()
```

8.5.3.5 setApple()

```
void nhf.core.Map.setApple (
            Point p)
```

8.5.3.6 setPowerUp()

```
void nhf.core.Map.setPowerUp (
            Point p,
            ObjectType ot)
```

8.5.4 Member Data Documentation

8.5.4.1 SIZE

```
final int nhf.core.Map.SIZE = 11  [static]
```

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/core/Map.java

8.6 nhf.gui.GamePanel.MenuPanel Class Reference

Public Member Functions

- MenuPanel (Image backgroundImg)
- void paintComponent (Graphics g)

8.6.1 Detailed Description

The panel displayed in the main menu

8.6.2 Constructor & Destructor Documentation

8.6.2.1 MenuPanel()

```
nhf.gui.GamePanel.MenuPanel.MenuPanel (
            Image backgroundImg)
```

8.6.3 Member Function Documentation

8.6.3.1 paintComponent()

```
void nhf.gui.GamePanel.MenuPanel.paintComponent (
            Graphics g)
```

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/gui/GamePanel.java

8.7 nhf.core.ObjectType Enum Reference

Public Attributes

- APPLE
- SPEED
- INVERTED
- MEGAAPPLE

8.7.1 Detailed Description

Types of objects that can appear on the map

8.7.2 Member Data Documentation

8.7.2.1 APPLE

```
nhf.core.ObjectType.APPLE
```

8.7.2.2 INVERTED

```
nhf.core.ObjectType.INVERTED
```

8.7.2.3 MEGAAPPLE

```
nhf.core.ObjectType.MEGAAPPLE
```

8.7.2.4 SPEED

```
nhf.core.ObjectType.SPEED
```

The documentation for this enum was generated from the following file:

- snakegame/src/main/java/nhf/core/ObjectType.java

8.8 nhf.gui.ScorePanel Class Reference

Public Member Functions

- ScorePanel (Image bg, GamePanel parentFrame)

Protected Member Functions

- void paintComponent (Graphics g)

8.8.1 Detailed Description

A panel that displays the high score table with a background image

8.8.2 Constructor & Destructor Documentation

8.8.2.1 ScorePanel()

```
nhf.gui.ScorePanel.ScorePanel (
            Image bg,
            GamePanel parentFrame)
```

Creates a ScorePanel with the given background image and parent frame

Parameters

bg	
parentFrame	

8.8.3 Member Function Documentation

8.8.3.1 paintComponent()

```
void nhf.gui.ScorePanel.paintComponent (
            Graphics g)  [protected]
```

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/gui/ScorePanel.java

8.9 nhf.core.Scores Class Reference

Classes

- class **Entry**

Public Member Functions

- void saveScore (String playerName, int score, String time, String result)
- List< Entry > getTopScores ()

8.9.1 Detailed Description

Handles saving and loading of high scores

8.9.2 Member Function Documentation

8.9.2.1 getTopScores()

```
List< Entry > nhf.core.Scores.getTopScores ()
```

Loads and returns the list of top scores from the scores.json file

Returns

list of top score entries

8.9.2.2 saveScore()

```
void nhf.core.Scores.saveScore (
            String playerName,
            int score,
            String time,
            String result)
```

Saves a new score entry to the scores.json file

Parameters

playerName	
score	
time	
result	

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/core/Scores.java

8.10 nhf.core.Snake Class Reference

Public Member Functions

- Snake ()
- void move (int dx, int dy, boolean appleEaten)
- void increaseSize (int dx, int dy, boolean appleEaten)
- Point getHead ()
- List< Point > getBody ()
- int getSize ()

8.10.1 Detailed Description

The snake controlled by the player

8.10.2 Constructor & Destructor Documentation

8.10.2.1 Snake()

```
nhf.core.Snake.Snake ()
```

8.10.3 Member Function Documentation

8.10.3.1 getBody()

```
List< Point > nhf.core.Snake.getBody ()
```

8.10.3.2 getHead()

```
Point nhf.core.Snake.getHead ()
```

8.10.3.3 getSize()

```
int nhf.core.Snake.getSize ()
```

8.10.3.4 increaseSize()

```
void nhf.core.Snake.increaseSize (
            int dx,
            int dy,
            boolean appleEaten)
```

8.10.3.5 move()

```
void nhf.core.Snake.move (
            int dx,
            int dy,
            boolean appleEaten)
```

Moves the snake in the given direction, growing it if an apple was eaten

Parameters

dx	
dy	
appleEaten	

The documentation for this class was generated from the following file:

- snakegame/src/main/java/nhf/core/Snake.java

Chapter 9

File Documentation

9.1 snakegame/src/main/java/nhf/core/Engine.java File Reference

```
import java.awt.Point;
import java.awt.event.KeyEvent;
import java.util.Random;
import javax.swing.JOptionPane;
import javax.swing.Timer;
import nhf.gui.GamePanel.InGamePanel;
```

Classes

- class nhf.core.Engine

Packages

- package nhf.core

9.2 snakegame/src/main/java/nhf/core/Map.java File Reference

```
import java.awt.Point;
```

Classes

- class nhf.core.Map

Packages

- package nhf.core

9.3 snakegame/src/main/java/nhf/core/ObjectType.java File Reference

Classes

- enum nhf.core.ObjectType

Packages

- package nhf.core

9.4 snakegame/src/main/java/nhf/core/Scores.java File Reference

```
import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import javax.json.Json;
import javax.json.JsonArray;
import javax.json.JsonArrayBuilder;
import javax.json.JsonObject;
import javax.json.JsonReader;
import javax.json.JsonWriter;
import javax.json.JsonWriterFactory;
import javax.json.stream.JsonGenerator;
```

Classes

- class nhf.core.Scores
- class **nhf.core.Scores.Entry**

Packages

- package nhf.core

9.5 snakegame/src/main/java/nhf/core/Snake.java File Reference

```
import java.awt.Point;
import java.util.LinkedList;
import java.util.List;
```

Classes

- class nhf.core.Snake

Packages

- package nhf.core

9.6 snakegame/src/main/java/nhf/gui/GamePanel.java File Reference

```
import javax.swing.Box;
import javax.swing.BoxLayout;
import javax.swing.ImageIcon;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JPanel;
import java.awt.Color;
import java.awt.Font;
import java.awt.Graphics;
import java.awt.Image;
import java.awt.Point;
import java.awt.event.ComponentAdapter;
import java.awt.event.ComponentEvent;
import java.awt.event.KeyAdapter;
import java.awt.event.KeyEvent;
import java.util.List;
import nhf.core.Engine;
import nhf.core.Map;
import nhf.core.Snake;
import nhf.core.ObjectType;
```

Classes

- class nhf.gui.GamePanel
- class nhf.gui.GamePanel.MenuPanel
- class nhf.gui.GamePanel.InGamePanel

Packages

- package nhf.gui

9.7 snakegame/src/main/java/nhf/gui/ScorePanel.java File Reference

```
import javax.swing.*;
import java.awt.*;
import java.util.List;
import javax.swing.table.DefaultTableCellRenderer;
import javax.swing.table.JTableHeader;
import nhf.core.Scores;
```

Classes

- class nhf.gui.ScorePanel

Packages

- package nhf.gui

9.8 snakegame/src/main/java/nhf/Main.java File Reference

```
import nhf.gui.*;
```

Classes

- class nhf.Main

Packages

- package nhf

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